Project: Worked examples
Title: Motor and actuator sizing
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1. Scope and method

Torque, speed and power a motor needs to move a load through a mechanism on a point-to-point cycle: the load reflected to the motor shaft, the acceleration from the motion profile, peak torque during acceleration, RMS torque over the cycle (thermal duty), peak speed, inertia ratio, and a check of a candidate motor's rated and peak torque and speed with a safety margin. A rotary load that runs continuously (a pump, a fan) is sized instead from its steady torque and speed and a run-up time.

WHAT IT CALCULATES	RELATION
Motor-to-load ratio of a linear mechanism, in metres per motor radian	$k = p/(2\pi\, i)$ (screw); $D/(2\, i)$ (belt); $D_p/(2\, i)$ (rack)
Load inertia reflected to the motor through a linear mechanism	$J_{refl} = m\, k^2/\eta$
Load torque at the motor through a linear mechanism	$T_{load} = F\, k/\eta$, with $F = m\, g\, (\sin\theta + \mu \cos\theta) + F_p$
Reflected inertia and torque of a rotary load through a gearbox	$J_{refl} = J_L/(i^2\, \eta)$; $T_{load} = T_L/(i\, \eta)$
Move profile, trapezoidal 1/3–1/3–1/3 (adjustable)	$v_{pk} = s/(t_{move} - t_{acc})$; $a = v_{pk}/t_{acc}$
Move profile, S-curve (sine acceleration, same peak speed)	$a_{pk} = (\pi/2)\, a_{trap}$
Torque to accelerate the total inertia, and the peak torque	$T_{acc} = J_{total}\, \alpha$; $T_{pk} = T_{acc} + T_{load}$
Alternating torque amplitude of a point-to-point cycle about T_{load}, for a shaft's fatigue	$T_{alt} = (T_{pk} - T_{dec})/2$ (0 in continuous duty). Torque behind the ratio on the output shaft of a rotary mechanism: $T_{out} = T \cdot i \cdot \eta$ (mean, alternating and peak), $n_{out} = n_{pk}/i$. Peak axial thrust of a ball screw on its support bearings: $F_{ax,pk} = F_{load} + m\, a_{pk}$ (0 for other mechanisms)
RMS (thermal) torque over the whole cycle, dwell included	$T_{rms} = \sqrt{(\Sigma\, T_i^2\, t_i / t_{cycle})}$
RMS mechanical power over the cycle, the thermal-equivalent continuous power a supply sees (regeneration counts as power; the instantaneous peak P_{pk} is not a continuous demand), trapezoidal profile	$P_{rms} = \omega_{pk}\, \sqrt{\{[t_{acc} \cdot (2/3)(T_{load}^2 + T_{acc}^2) + T_{load}^2\, t_{const}]/t_{cycle}\}}$; for the S-curve the acceleration term becomes $t_{acc} \cdot [(3/4)\, T_{load}^2 + (5/16)\, T_{acc}^2]$
Continuous rotary duty (a pump or fan), speed at the motor	$\omega = 2\pi\, n_L\, i/60$; $n_{pk} = n_L\, i$
Continuous rotary duty, load torque and RMS torque	$T_{load} = T_L/(i\, \eta)$; $T_{rms} = T_{load}$
Peak torque during the run-up, with full load throughout (conservative)	$T_{pk} = J_{total}\, \omega/t_{start} + T_{load}$

WHAT IT CALCULATES	RELATION
Continuous power	$P_{cont} = T_{load}\,\omega$; $P_{rms} = P_{cont}$
Rated shaft power of the candidate motor, from its rated torque and its rated speed n_{rated} (not the running speed), the shaft power its supply must be sized for; without n_{rated} there is no P_{rated} and the sheet warns	$P_{rated} = T_{rated} \cdot 2\pi\, n_{rated}/60$

Basis: Load torque, inertia referred to the motor, duty cycles[1](Ch. on motor and drive selection); Reflected inertia, RMS torque, inertia ratio practice[2].

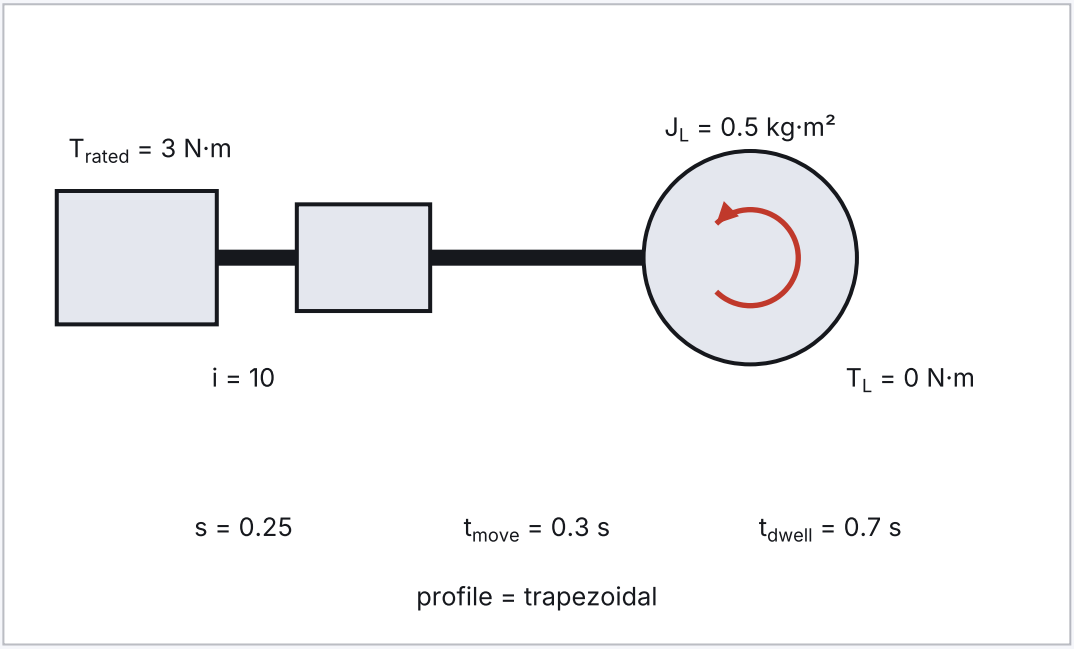


Figure 1 — A motor through a gearbox to a rotary load of inertia J_L and torque T_L. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mechanism	Mechanism	gearbox rotary		assumption: Sizing convention.	any
i	Gear ratio i (motor : output, 1 = none)	10		spec: Gearbox datasheet PG-10: 10:1, 85 %	1 … 1000
η	Mechanism efficiency (overall)	0.85		spec: Gearbox datasheet PG-10: 10:1, 85 %	0.3 … 1
J_{mech}	Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)	0	kg·m²	assumption: Sizing convention.	0 … 10 kg·m²
μ	Friction coefficient (guide / bearing)	0.1		default	0 … 1
θ	Incline from horizontal (+ moving uphill)	0	°	default	-90 … 90 °

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
F_p	Process force opposing the move (linear)	0	N	default	0 … 1000000 N
J_L	Load inertia at the output (rotary mechanisms)	0.5	kg·m²	spec: S1-style inertia calc of the 600 mm table with fixtures: 0.5 kg·m²	0 … 10000 kg·m²
T_L	Constant load torque at the output (rotary)	0	N·m	assumption: Sizing convention.	0 … 100000 N·m
$duty_{mode}$	Duty (point-to-point move cycle, or continuous rotary running with a run-up)	point-to-point		default	any
profile	Motion profile	trapezoidal		assumption: Sizing convention.	any
s	Move distance (linear: m; rotary: output revolutions)	0.25		spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.0001 … 10000
t_{move}	Move time	0.3	s	spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.01 … 3600 s
f_{acc}	Acceleration (and deceleration) fraction of the move time	0.333333		spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0.05 … 0.5
t_{dwell}	Dwell between moves	0.7	s	spec: Cycle specification CS-7: 90° index in 0.3 s, 0.7 s dwell	0 … 3600 s
J_{motor}	Candidate motor rotor inertia	0.0005	kg·m²	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	1e-7 … 10 kg·m²
T_{rated}	Candidate motor continuous (rated) torque	3	N·m	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	0.01 … 100000 N·m
T_{peak}	Candidate motor peak torque	9	N·m	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	0.01 … 100000 N·m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
n_{max}	Candidate motor maximum speed	3000	rpm	spec: Motor datasheet SM-060: 5.0e-4 kg·m², 3.0 / 9.0 N·m, 3000 rpm	10 … 100000 rpm
SF	Safety margin on torque	1.25		assumption: Sizing convention.	1 … 3
$J_{ratio,max}$	Inertia ratio limit (load : motor), warning	10		assumption: Sizing convention.	1 … 100

3. Assumptions

A1. Rigid transmission; the mechanism efficiency applies to the driving direction and is also used in the deceleration phase (conservative for peak, slightly pessimistic for RMS). (Servo-sizing practice.)
A2. The friction force is Coulomb friction on the moved mass; viscous and breakaway friction are not modelled unless included in F_p. (Load model.)
A3. Continuous duty: the load torque is constant at the stated speed and the motor accelerates the whole inertia linearly to speed in t_{start} while carrying the full load torque (conservative); the move distance, move time, acceleration fraction, dwell and profile are ignored. (Run-up idealisation for pumps and fans.)
A4. The S-curve profile has sine-shaped acceleration over the same acceleration time, so its peak acceleration is π/2 times the trapezoidal value. (Jerk-limited profile idealisation.)
A5. mechanism (Mechanism) is taken as gearbox rotary. (Sizing convention.)
A6. J_{mech} (Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)) is taken as 0 kg*m^2. (Sizing convention.)
A7. T_L (Constant load torque at the output (rotary)) is taken as 0 N*m. (Sizing convention.)
A8. profile (Motion profile) is taken as trapezoidal. (Sizing convention.)
A9. SF (Safety margin on torque) is taken as 1.25. (Sizing convention.)
A10. $J_{ratio,max}$ (Inertia ratio limit (load : motor), warning) is taken as 10. (Sizing convention.)

4. Working

[21] k = 1 / i
= 1 / 10
= 0.1
Output radians per motor radian
[22] T_{load} = T_L / (i * η)
= 0 N·m / (10 × 0.85)
= 0 N·m
Load torque at the motor shaft
[23] J_{refl} = J_L / (i^2 * η)
= 0.5 kg·m² / (10^2 × 0.85)
= 0.0058824 kg·m²
Load inertia reflected to the motor shaft
[24] $θ_m$ = s * 2 * pi * i
= 0.25 × 2 × π × 10
= 15.708 rad
Motor angle for the move
[25] $ω_{pk}$ = $θ_m$ / (t_{move} * (1 - f_{acc}))
= 15.708 rad / (0.3 s × (1 - 0.33333))
= 78.54 rad/s
Peak motor speed (trapezoid)
[26] $α_{trap}$ = $ω_{pk}$ / (f_{acc} * t_{move})
= 78.54 rad/s / (0.33333 × 0.3 s)
= 785.4 rad/s²

Trapezoidal angular acceleration at the motor

$$[28]\ \alpha_{pk} = \alpha_{trap}$$
$$= 785.4\ \text{rad/s}^2$$
$$= 785.4\ \text{rad/s}^2$$

Peak acceleration

$$[29]\ \alpha_{pk} = \alpha_{pk}$$
$$= 785.4\ \text{rad/s}^2$$
$$= 785.4\ \text{rad/s}^2$$

Peak motor angular acceleration

$$[30]\ J_{total} = J_{motor} + J_{mech} + J_{refl}$$
$$= 0.0005\ \text{kg·m}^2 + 0\ \text{kg·m}^2 + 0.0058824\ \text{kg·m}^2$$
$$= 0.0063824\ \text{kg·m}^2$$

Total inertia at the motor shaft

$$[31]\ T_{acc} = J_{total} * \alpha_{pk}$$
$$= 0.0063824\ \text{kg·m}^2 \times 785.4\ \text{rad/s}^2$$
$$= 5.0127\ \text{N·m}$$

Acceleration torque

$$[32]\ T_{pk} = T_{acc} + T_{load}$$
$$= 5.0127\ \text{N·m} + 0\ \text{N·m}$$
$$= 5.0127\ \text{N·m}$$

Peak torque (accelerating against the load)

$$[33]\ T_{dec} = -T_{acc} + T_{load}$$
$$= -5.0127\ \text{N·m} + 0\ \text{N·m}$$
$$= -5.0127\ \text{N·m}$$

Torque during deceleration

$$[34]\ T_{alt} = (T_{pk} - T_{dec}) / 2$$
$$= (5.0127\ \text{N·m} - (-5.0127\ \text{N·m})) / 2$$
$$= 5.0127\ \text{N·m}$$

Alternating torque amplitude of the cycle about T_{load}

$$[35]\ F_{ax,pk} = 0\ \text{N}$$
$$= 0\ \text{N}$$

Peak axial thrust on the bearings (none for this mechanism)

$$[36]\ n_{pk} = \omega_{pk}$$
$$= 78.54\ \text{rad/s}$$
$$= 750\ \text{rpm}$$

Peak motor speed

$$[37]\ T_{out,m} = T_{load} * i * \eta$$
$$= 0\ \text{N·m} \times 10 \times 0.85$$
$$= 0\ \text{N·m}$$

Mean torque on the output shaft (the constant load torque)

$$[38]\ T_{out,a} = T_{alt} * i * \eta$$
$$= 5.0127\ \text{N·m} \times 10 \times 0.85$$
$$= 42.608\ \text{N·m}$$

Alternating torque amplitude on the output shaft

$$[39]\ T_{out,pk} = T_{pk} * i * \eta$$
$$= 5.0127\ \text{N·m} \times 10 \times 0.85$$
$$= 42.608\ \text{N·m}$$

Peak torque on the output shaft (the motor rotor's inertia torque is counted, so conservative)

$$[40]\ n_{out} = n_{pk} / i$$
$$= 750\ \text{rpm} / 10$$
$$= 75\ \text{rpm}$$

Output shaft speed

$$[41]\ P_{pk} = T_{pk} * \omega_{pk}$$
$$= 5.0127\ \text{N·m} \times 78.54\ \text{rad/s}$$
$$= 0.3937\ \text{kW}$$

Peak mechanical power

[42] $t_{acc} = f_{acc} * t_{move}$

$= 0.33333 \times 0.3\ s$

$= 0.1\ s$

Acceleration time

[43] $t_{const} = t_{move} - 2 * t_{acc}$

$= 0.3\ s - 2 \times 0.1\ s$

$= 0.1\ s$

Constant-velocity time

[44] $t_{cycle} = t_{move} + t_{dwell}$

$= 0.3\ s + 0.7\ s$

$= 1\ s$

Cycle time

[45] $P_{rms} = \omega_{pk} * \mathrm{sqrt}((t_{acc} * (2 / 3) * (T_{load}\text{^}2 + T_{acc}\text{^}2) + T_{load}\text{^}2 * t_{const}) / t_{cycle})$

$= 78.54\ rad/s \times \mathrm{sqrt}((0.1\ s \times 2 / 3 \times ((0\ N{\cdot}m)\text{^}2 + (5.0127\ N{\cdot}m)\text{^}2) + (0\ N{\cdot}m)\text{^}2 \times 0.1\ s) / 1\ s$

$= 0.10165\ kW$

RMS mechanical power over the cycle (thermal-equivalent continuous power)

[46] $T_{rms} = \mathrm{sqrt}((T_{pk}\text{^}2 * t_{acc} + T_{load}\text{^}2 * t_{const} + T_{dec}\text{^}2 * t_{acc}) / t_{cycle})$

$= \mathrm{sqrt}(((5.0127\ N{\cdot}m)\text{^}2 \times 0.1\ s + (0\ N{\cdot}m)\text{^}2 \times 0.1\ s + (-5.0127\ N{\cdot}m)\text{^}2 \times 0.1\ s) / 1\ s)$

$= 2.2417\ N{\cdot}m$

RMS torque over the cycle (thermal)

[47] $J_{ratio} = (J_{refl} + J_{mech}) / J_{motor}$

$= (0.0058824\ kg{\cdot}m^2 + 0\ kg{\cdot}m^2) / 0.0005\ kg{\cdot}m^2$

$= 11.765$

Inertia ratio (load : motor)

[48] $u_{peak} = SF * T_{pk} / T_{peak}$

$= 1.25 \times 5.0127\ N{\cdot}m / 9\ N{\cdot}m$

$= 0.69621$

Peak torque utilisation with safety margin

[49] $u_{rms} = SF * T_{rms} / T_{rated}$

$= 1.25 \times 2.2417\ N{\cdot}m / 3\ N{\cdot}m$

$= 0.93406$

Continuous torque utilisation with safety margin

[50] $u_{speed} = n_{pk} / n_{max}$

$= 750\ rpm / 3000\ rpm$

$= 0.25$

Speed utilisation

[51] $duty = T_{rms} / T_{rated}$

$= 2.2417\ N{\cdot}m / 3\ N{\cdot}m$

$= 0.74725$

Thermal duty (without margin)

5. Results

SYMBOL	VALUE	DESCRIPTION
k	0.1	Transmission constant
T_{load}	0 N·m	Load torque at motor
J_{refl}	0.0058824 kg·m²	Reflected load inertia
J_{total}	0.0063824 kg·m²	Total inertia
θ_m	15.708 rad	Motor angle
a_{pk}	785.4 rad/s²	Peak acceleration
α_{pk}	785.4 rad/s²	Motor angular acceleration

SYMBOL	VALUE	DESCRIPTION
T_{acc}	5.0127 N·m	Acceleration torque
T_{pk}	5.0127 N·m	Peak torque
T_{dec}	-5.0127 N·m	Deceleration torque
T_{alt}	5.0127 N·m	Alternating torque amplitude
$F_{ax,pk}$	0 N	Peak axial thrust on a screw's bearings
T_{rms}	2.2417 N·m	RMS torque
n_{pk}	750 rpm	Peak speed
P_{pk}	0.3937 kW	Peak power
P_{rms}	0.10165 kW	RMS power (continuous equivalent)
J_{ratio}	11.765	Inertia ratio
duty	0.74725	Thermal duty
u_{peak}	0.69621	Peak utilisation
u_{rms}	0.93406	Continuous utilisation
u_{speed}	0.25	Speed utilisation
$T_{out,m}$	0 N·m	Output shaft mean torque
$T_{out,a}$	42.608 N·m	Output shaft alternating torque
$T_{out,pk}$	42.608 N·m	Output shaft peak torque
n_{out}	75 rpm	Output shaft speed

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	RMS torque $SF \cdot T_{rms} \le T_{rated}$	u_{rms} = 0.93406	$u_{rms} \le 1$	0.934	
PASS	Peak torque $SF \cdot T_{pk} \le T_{peak}$	u_{peak} = 0.69621	$u_{peak} \le 1$	0.696	
PASS	Speed $n_{pk} \le n_{max}$	u_{speed} = 0.25	$u_{speed} \le 1$	0.25	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: J_{motor}, T_{rated}, T_{peak}, n_{max} (Candidate motor rating). Enter each as one so its source, edition and licence travel with it.
WARNING — input.mu: μ (Friction coefficient (guide / bearing)) was not entered; the template default 0.1 is used.
WARNING — input.theta: θ (Incline from horizontal (+ moving uphill)) was not entered; the template default 0 deg is used.
WARNING — input.F_p: F_p (Process force opposing the move (linear)) was not entered; the template default 0 N is used.
WARNING — input.duty_mode: $duty_{mode}$ (Duty (point-to-point move cycle, or continuous rotary running with a run-up)) was not entered; the template default 'point-to-point' is used.
WARNING — inertia-ratio: J_{motor}: the inertia ratio 11.8 exceeds 10.

WARNING — overhauling: T_{dec}: the motor brakes during deceleration (negative torque); the drive must absorb the regenerated energy.

8. Verification

```
Template elec.motor-sizing@1.6.0 Content hash 4ea5e25d0b648caa Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

9. References

1. Hughes A, Drury B. Electric Motors and Drives: Fundamentals, Types and Applications. 5th ed. Newnes; 2019.
2. Kollmorgen; Parker Hannifin. Servo motor sizing guides [manufacturer technical literature].